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Visvesvaraya Technological University

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Project Report

On

"Smart Coco: Real-Time Monitoring and Optimization of Coconut Farming"

Submitted in partial fulfilment for the award of degree
of Bachelor of Engineering

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a bonafide students of VII Semester Electronics & Communication Engineering, Atria Institute of Technology, **Bengaluru** in partial fulfillment for the award of Bachelor of Engineering in Visvesvaraya Technological University, Belagavi during the year **2025-26**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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ABSTRACT

Agriculture is a crucial sector that directly impacts food production and economic stability. With technological advancements, smart agricultural systems can help farmers optimize resources and improve productivity. This project focuses on real-time soil analysis and coconut disease detection using image processing. The soil analysis module integrates NPK, pH, and moisture sensors to measure essential soil parameters. The collected data is displayed on an OLED/LCD screen and stored in cloud storage for remote access and long-term monitoring. Simultaneously, the image processing where the captured images of coconut leaves are processed using CNN techniques to detect the disease accurately. By providing a cost-effective and remotely accessible solution, this system enables farmers to make informed decisions regarding soil health and yield estimation, ultimately promoting sustainable and precision farming practices.

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CHAPTER 1

INTRODUCTION

Agricultural innovation is paramount to establishing sustainability, addressing food needs in a developing population, and establishing economic stability. Climate change, soil erosion, and water shortages present challenges that conventional agriculture is not enough to fulfill the growing needs for these. With the advancement of technologies such as precision farming, artificial intelligence, and sensor-based monitoring, agriculture is being revolutionized by minimizing its impact on the environment and encouraging sustainable food production. By adopting these innovations, farmers can optimize resource use, improve crop quality, and reduce vulnerability to unpredictable challenges.

Coconut cultivation remains a vital source of livelihood for thousands of farmers worldwide, significantly contributing to the economy and generating export income. Each and every element of the coconut the husk, shell, and fruit yielding healthy milk, oil, and water is a worthy justification for its reputation as the "tree of life." Aside from economic advantages, coconut trees contribute to soil conservation, carbon sequestration, and ecological rehabilitation of deforested lands. Coconut cultivation promotes sustainable development through the encouragement of food security, industrial growth, and employment, especially in the countryside.

Although critical to the economy, coconut production struggles with productivity loss from pests such as rhinoceros beetles and red palm weevils, and diseases such as root wilt and bud rot, which cut yields dramatically. Climate change also aggravates such issues through unseasonal rains, dry spells, and intense temperatures that throw the coconut growth cycle off track. To solve these problems, new technologies such as real-time soil monitoring, early disease detection using image processing, and automation of management systems provide feasible solutions. These technologies assist in the optimization of the use of inputs, early disease detection, and decision-making recommendations, thereby guaranteeing sustainable yield gains and farmers' livelihoods.

Real-time monitoring of soil is an important aspect of precision agriculture, constantly measuring essential soil parameters including moisture, pH, temperature, and nutrient content to intervene in a timely manner to ensure soil health. Early detection and classification of coconut diseases are possible through image processing methods that help farmers avoid extensive damage. A simple interface provides easy access to such information and management recommendations, assisting farmers in

decision-making. Collectively, these technologies encourage sustainable agriculture by maximizing the utilization of resources, minimizing environmental footprint, and maximizing coconut yields through accurate, data-driven management practices.

Coconut cultivation is crucial for the economy and livelihoods of thousands of farmers worldwide. However, it faces challenges like pests, diseases, and climate change, which threaten productivity and sustainability. Innovative technologies like real-time soil monitoring, image processing for disease detection, and automation can help optimize resource use, detect issues early, and support data-driven decision-making. By adopting these solutions, farmers can improve yields, reduce environmental impact, and ensure sustainable coconut production.

Sustainable coconut cultivation is essential for maintaining the livelihoods of farmers and the health of the environment. With the global demand for coconut products increasing, it's crucial to adopt practices that prioritize sustainability and minimize harm to the ecosystem. By leveraging technologies like precision agriculture and data-driven decision-making, farmers can optimize their yields while reducing their environmental footprint.

Technology is transforming the coconut farming industry, enabling farmers to make data-driven decisions and optimize their practices. From real-time soil monitoring to image processing for disease detection, these innovations are helping farmers to improve yields, reduce waste, and promote sustainability. By embracing these technologies, farmers can stay competitive in a rapidly changing market and ensure the long-term viability of their crops.

The incorporation of IoT devices, cloud platforms enhances coconut farming efficiency even more, by enabling farmers to supervise and control field conditions remotely. These systems enable the farmer to receive instant alerts on existing conditions and track long-term soil and crop trends and make changes in nutrient supply without human intervention. Such intelligent farming solutions reduce labor dependency and ensure resource application with accuracy, leading to lower production costs and wastage of resources. If integrated with predictive analytics and weather forecasting tools, farmers can anticipate adverse climatic events, plan field activities in advance, and safeguard their plantations from potential losses. Therefore, this new wave of digital agriculture makes the coconut cultivation ecosystem more resilient and sustainable, equipping farmers with the needed tools to overcome traditional challenges and ensure consistent, quality yields.

1.1 Motivation

Smart Coco is not just a technological innovation; it's a lifeline for many coconut farmers, especially those in rural areas. For example, Coco-Bot, a part of this smart farming revolution, has helped rural women earn around ₹18,000–₹20,000 a month by making coconut farming more efficient and less labour-intensive. This technology has transformed the traditional hardships of coconut farming climbing tall trees, unpredictable yields, and manual labour into a streamlined, manageable process where farmers can rely on data and automation. It has brought new hope and dignity to these communities, empowering them with better income and safer working conditions. Stories like this highlight how Smart Coco is making farming more humane and sustainable by bridging the gap between modern technology and the everyday lives of farmers.

With Smart Coco, this farmer now receives precise guidance right on their phone—knowing exactly when to water, how to spot diseases early, or when the coconuts are ready to pick. It is also making the hard work accessible to women and others who couldn't join traditional farming before. By turning difficult manual work into smart, guided farming, Smart Coco brings peace of mind, better income, and improved safety, helping families thrive and feel hopeful about their future. This is technology with a human heart, empowering farmers every step of the way.

1.2 Scope of work

The scope of work for "Smart Coco: Real-Time Monitoring and Optimization of Coconut Farming" involves developing and deploying advanced technologies such as IoT devices, AI-powered platforms, and machine learning to help coconut farmers monitor crop health, track coconut maturity, and make data-driven decisions. The goal is to enable precise detection of coconut maturity, optimize harvest timing, improve yield, conserve resources like water and fertilizer, and make the overall process more efficient and sustainable for farmers. This project is designed to empower coconut growers with actionable insights, ensure better quality produce, and support sustainable agricultural practices through cost-effective, real-time digital solutions.

Smart Coco is a project designed to bring simple, practical technology into coconut farming to make the entire process easier, smarter, and more efficient for farmers. Instead of relying on guesswork or traditional methods, the system uses IoT devices placed around the farm to constantly monitor soil moisture, temperature, humidity, nutrient levels, and overall tree health. These devices act like digital assistants, giving farmers real-time information about what their trees need. Along with this, AI and machine learning tools analyse images of coconuts to accurately determine their maturity level, helping farmers know the perfect time to harvest and avoid picking too early or too late. This leads to better-quality produce and higher market prices.

The platform also gives guidance on water usage and fertilizer application, ensuring resources are used wisely and sustainably. Farmers can view all this information on an easy-to-use app or dashboard, where they receive alerts, suggestions, and a clear understanding of farm conditions. Over time, this data helps them make smarter decisions, improve yield, reduce wastage, and lower costs. Overall, Smart Coco empowers coconut farmers with timely insights, improves productivity, supports sustainable practices, and makes farm management more accurate and stress-free.

1.3 Problem Statement

- Agriculturalists find it challenging to observe soil condition and identify crop disease in real time.
- Slow detection results in delayed response, lowered production of crops.
- Integrated, simple systems are not available for farmers.
- Specific, accurate disease detection of crop, particularly in coconut, is a critical need.

1.4 Objectives

- To develop a real-time soil monitoring system.
- To implement an image processing-based coconut disease detection system.
- To provide a user-friendly interface for farmers to monitor soil health.
- To suggest the management techniques for the detected disease.

CHAPTER 2

LITERATURE SURVEY

[1] Title: "Coconut Disease Prediction System Using Image Processing and Deep Learning Techniques"

Author: Dhapitha Nesarajan; Lokini Kunalan; Mithun Logeswaran, 2020 IEEE 4th International Conference on Image Processing, Applications and Systems (IPAS).

Contributions and Conclusions:

The research article "Coconut Disease Prediction System Using Image Processing and Deep Learning Techniques" demonstrates a solution implemented using Android that identifies pest infestation and nutrient stresses in coconut plants based on image processing and machine learning. The system photographs the leaves of coconut, pre-processes and separates them, and then identifies color and shape-related features to classify the disease based on SVM and CNN models. EfficientNetB0 model had the best accuracy of 93.72% to identify nutrient deficiencies. The system also recommends suitable pesticides or fertilizers and has a feature with OpenCV for monitoring the recovery of plants with the hope of helping farmers effectively enhance productivity and disease management.

[2] Title: "IoT-Based Systems for Soil Nutrients Assessment in Horticulture"

Author: Stefan Postolache, Pedro Sebastião, and Francisco Cercas, (2023). IoT-Based Systems for Soil Nutrients Assessment in Horticulture. Sensors, 23*(403)*.

Contributions and Conclusions:

The project presents an IoT-based system for soil nutrient assessment in horticulture. It highlights the importance of real-time monitoring of soil nutrients (N, P, K), moisture, pH, and electrical conductivity to enhance soil fertility management. Traditional soil analysis methods, while accurate, are costly and time-consuming. In contrast, the proposed system uses smart sensor nodes with wireless data transmission to provide real-time insights. The system's ability to integrate soil and air parameters ensures more accurate interpretations, ultimately aiding better irrigation and fertilizer management. The experimental results from the Tropical Botanical Garden in Lisbon validate the system's effectiveness in tracking soil parameter variations across different sites.

[3] Title: "Disease and pest infection detection in coconut tree through deep learning techniques"

Author: Piyush Singh [a], Abhishek Verma [a], John Sahaya Rani Alex, Computers and Electronics in Agriculture, 182, 105986. (2021)

Contributions and Conclusions:

The research paper "Disease and pest infection detection in coconut tree through deep learning techniques" presents an automatic system on the basis of deep learning to identify diseases and infestations by pests-i.e., stem bleeding, leaf blight, and red palm weevil infestation—of coconut trees. This study collected a data set of 1,564 images from Tamil Nadu's farms, up to around 39,000 images using image processing techniques. Various segmentation methods were tried, and K-means clustering proved to perform the best. An in-house 2D Convolutional Neural Network (CNN) was designed to achieve a high validation accuracy of 96.94% from segmented images, whereas pre-trained models like MobileNet and InceptionResNetV2 were used to achieve accuracy, employing transfer learning. The MobileNet model was extended to be used in a simple-to-use web application built using Flask, providing real-time upload of images and instant classification of diseases and suggested remedies. The approach provides an accessible, usable solution for farmers, facilitating early diagnosis and timely treatment, with further developments planned to provide cover over additional diseases and assess infection severity.

[4] Title: "A Crop Recommendation System Based on Nutrients and Environmental Factors Using Machine Learning Models and IoT"

Author: A Chauhan, A Tsundru, K Praveen, S Tokala, K Hajarathaiah and M K Endur, A review. Circular Agricultural Systems, 2(7). (2022)

Contributions and Conclusions:

The research paper proposes a machine learning-based crop recommendation system that considers soil nutrients, pH, temperature, and humidity to suggest optimal crops. The model uses algorithms like Random Forest to make accurate predictions, aiming to maximize agricultural productivity and sustainability. This system helps farmers make informed decisions to enhance crop yield results.

[5] Title: "Review on integrated pest management of coconut crop"

Author: Abhishek TS, SA Dwivedi, International Journal of Entomology Research, International Journal of Entomology Research, 6(3), 115–120. (2021)

Contributions and Conclusions:

The article reviews integrated pest management (IPM) strategies for coconut crops, highlighting major pests and sustainable control methods. Early detection, monitoring, and farmer awareness are crucial for successful IPM, minimizing crop loss and improving coconut productivity.

[6] Title: "Coconut Disease Prediction System Using Image Processing and Deep Learning Techniques"

Author: 2020 IEEE 4th International Conference on Image Processing, Applications and Systems (IPAS)

Contributions and Conclusions:

The article presents a coconut disease prediction system using image processing and deep learning, identifying pest infestations and nutrient stresses via Android app. It recommends suitable pesticides/fertilizers and monitors plant recovery, aiding farmers in enhancing productivity and disease management.

[7] Title: "Drone-mounted remote-controlled arm for monitoring and precision spraying coconut rhinoceros beetle infestations"

Author: Grianggai Samseemoung , Manusak Janthong , Weerayuth Promjan ,Volume 8, August 2024, 100438

Contributions and Conclusions:

A low-cost drone system with a remote-controlled sprayer arm enables real-time monitoring and precision pesticide application against coconut rhinoceros beetle infestations. The system uses a modified camera and custom image processing algorithms to detect infestation points and control spray operations effectively. Field tests showed reduced labor, chemical usage, and health risks while maintaining spray

precision under varying wind conditions.

8] Title: "Enhanced Coconut Yield Prediction Using Internet of Things and Deep Learning: A Bi-Directional Long Short-Term Memory Lévy Flight and Seagull Optimization Algorithm Approach"

Author: Rami N. Alkhawaji, Suhail H. Serbaya, Siraj Zahran, Applied Sciences, Volume 14, Issue 17, A Bi-Directional Long Short-Term Memory Lévy Flight and Seagull Optimization Algorithm Approach. .(2024)

Contributions and Conclusions:

The article presents an IoT-based deep learning model (BILSTM-LFSOA) for precise coconut yield prediction, using real-time environmental data. The model captures temporal dependencies and optimizes hyperparameters for high accuracy (98.96%). This approach provides valuable insights for farmers, policymakers, and stakeholders to optimize agricultural productivity and sustainability.

[9] Title: "Modeling and Forecasting of Coconut Area, Production, and Productivity Using a Time Series Model"

Author: BIO Web of Conference 97,(2024) https://doi.org/10.1051/bioconf/20249700113

ISCKU 2024

Contributions and Conclusions:

The paper uses the ARIMA model to analyze and forecast trends in coconut cultivation area and production in India. Based on historical data from 1950-51 to 2011-12, the research identifies growth patterns and selects suitable ARIMA models for predictions. The study projects a continued rise in both area and production, providing valuable insights for policymakers and farm planners.

CHAPTER 3

METHODOLOGY

The project follows a structured methodology consisting of three key phases:

- Soil analysis.
- Image processing for coconut disease detection.
- System Integration Implying the Disease Management Methods.

1.Soil Analysis:

The monitoring of soil during the soil monitoring phase employs sensors to gauge vital parameters like moisture, pH, temperature, and NPK values. Sensors are interfaced with a NodeMCU that captures data and sends it to a cloud platform through Wi-Fi connectivity. This system facilitates real-time monitoring and remote access to soil health data in a web or mobile interface. With the ongoing flow of soil data, farmers can make timely decisions regarding fertilizer application and irrigation.

2.Image Processing for Detection of Coconut Disease:

Coconut leaf images captured by mobile phones are processed using JSON with OpenCV. A Convolutional Neural Network (CNN) then classifies these segmented regions into specific disease types, with results stored in the cloud for generating reports and farmer alerts.

JSON uses OpenCV to process coconut leaf images taken from mobile phones. OpenCV is an open-source library of computer vision; it contains routines for image preprocessing. JSON's simplicity and vast support for scientific computing make it the perfect choice to implement image-processing pipelines and integrate machine learning models. A CNN then classifies such segmented regions into specific disease types with high accuracy based on learned visual patterns. Results are sent to the cloud, where automated systems generate detailed reports and push timely alerts to farmers for quick intervention, thus improving crop health management.

3.System Integration Implying the Disease Management Methods:

In this stage, the disease detection results and soil data are merged into a common cloud-based dashboard that can be accessed through mobile or web. Farmers can see real-time soil health and are notified of any likely coconut disease. The system also suggests management practices based on the information, such as the amount and type of fertilizer, irrigation planning, and organic treatment methods for detected diseases. This convergence provides an in-depth decision-making support system that enhances yield, minimizes losses, and facilitates sustainable coconut cultivation

This project integrates real-time soil measurement, high-level image processing for disease detection, and an integrated management system for sustainable coconut cultivation. The soil sensors on a ESP32 continuously measure important parameters and transfer the data to the cloud, facilitating remote access and timely decision-making. Soil and disease data are consolidated into a cloud-based, easy-to-use platform that gives farmers actionable advice for disease management and soil management. It gives farmers accurate, data-driven information to optimize crop health and maximize yield with efficient usage.

Integration of these technologies would involve a completely connected ecosystem in which data flows smoothly right from field sensors and mobile disease-detection modules to the cloud platform. With an ESP32-based system, communication will be low-power and cost-effective, thus enabling farmers in rural areas to acquire critical information without any expensive infrastructure. It combines soil analytics with disease classification data for automatically finding the stress conditions and predicting outbreaks before it gets out of hand.

The dashboard can also generate weekly/monthly reports, tracking improvements and planning long-term management strategies. Over time, it learns from the collected data, making more accurate recommendations possible via machine-learning models. Such intelligent automation reduces not just manual workload but also supports the cause of sustainable farming by preventing overuse of water, fertilizers, and pesticides. Overall, the integrated system will behave like a smart assistant for farmers to build better resilience and productivity in coconut cultivation.

3.1 Block Diagram

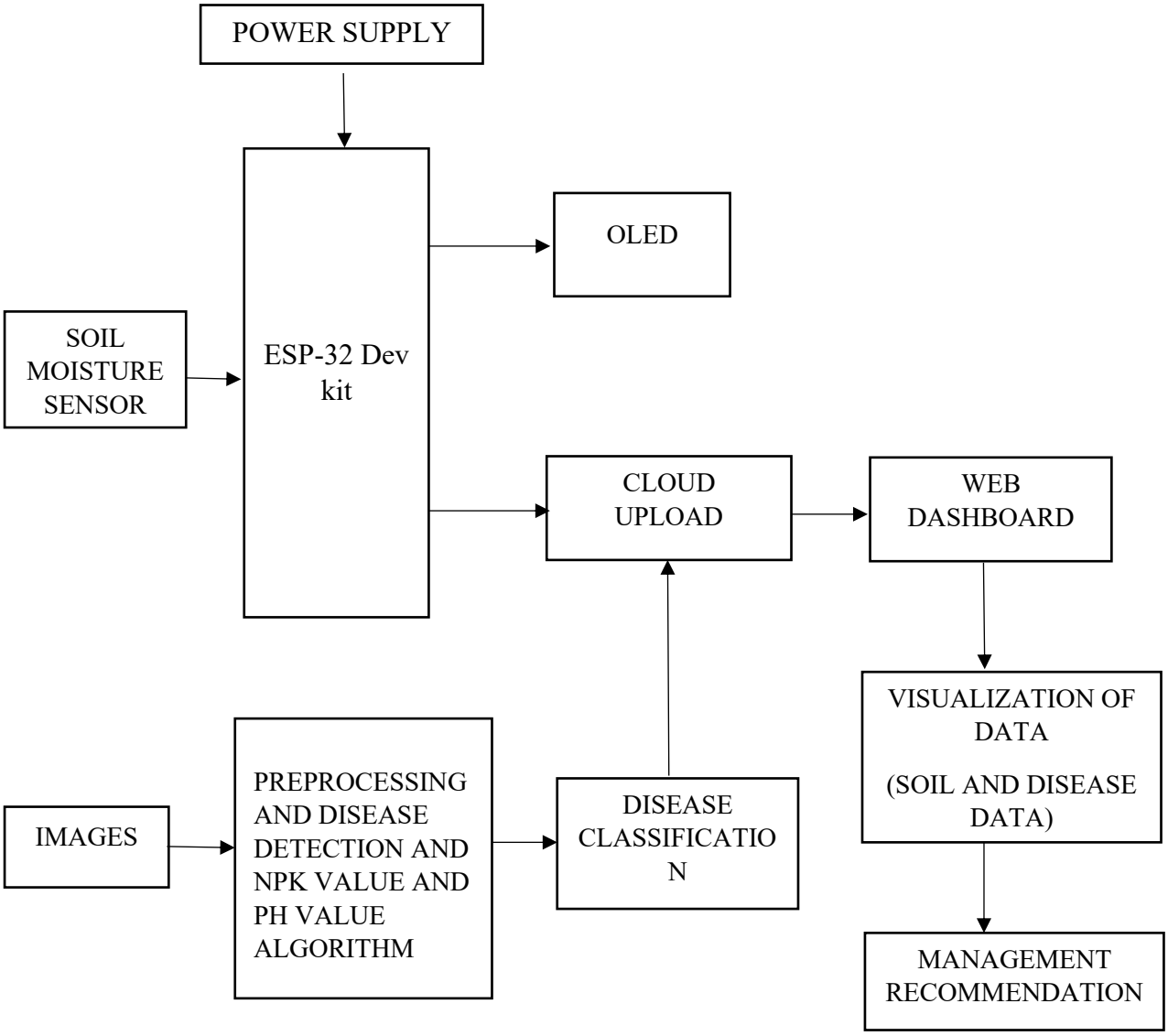

Fig 3.1 Block diagram of Smart Coco Monitoring and Farming

- This block diagram outlines a smart agricultural monitoring and management system.

1.Power Supply

The power supply is the backbone of any smart agricultural system, ensuring that every sensor, communication module, and microcontroller is provided with stable and uninterrupted electricity. A reliable power source is required because fluctuations may result in incorrect readings from sensors or failures in critical monitoring systems. Supply designs can be based on a battery, adapter, or even solar powered for field applications. It ensures that the voltage and current level is maintained constantly to keep the overall functioning smooth and continuous.

2.ESP-32 Dev Kit

The ESP-32 acts as the central processing unit, organizing all activities of the system and serving as the core controller. It collects data from sensors, performs necessary calculations, and exchanges data between different hardware components of the system. Having Wi-Fi and Bluetooth integrated into one module, ESP-32 supports hassle-free data transfer to cloud platforms or dashboards. It also performs onboard tasks like showing results on the OLED display, executing simple logic, and maintaining accuracy with timely handling of data.

3.Soil Moisture Sensor

The soil moisture sensor is used to monitor the water content within the soil; as a result, it is useful in assessing when irrigation should be applied. It continuously sends values to the ESP-32, which reinterprets the values to comprehend the hydration status of the soil. With this, overwatering and drought, which have adverse effects on crop growth, will be eliminated. The sensor contributes in managing the waste of water and retaining good health in the development of the plants.

4.OLED Display

An OLED display provides immediately available visual access to the user with real-time readings from the soil moisture status to system status updates. This small but powerfully embedded screen ensures that farmers or technicians can check field conditions in a glance without the use of any external device. High contrast and low power make this display suitable for outdoor agricultural environments. The display acts as a quick status indicator, improving usability and providing instant feedback.

5.Images and Preprocessing

Image preprocessing, which generally involves plant leaves, aims at enhancing the quality and clarity of the captured images. Noise removal, resizing, color enhancement, or segmentation may be involved to bring out the key features. Proper preprocessing will ensure further analysis for disease detection and nutrient estimation is accurate. It also reduces computational load by removing unwanted elements from the image. This stage is very vital for reliable and consistent outcomes in automated agricultural systems.

6.Disease Detection & Classification Algorithm

The symptoms of discoloration, spots, or texture changes are identified in the processed images by the disease detection algorithm. Using machine-learning or deep-learning models, the algorithm classifies the type of disease affecting the plant. This helps farmers with early detection of problems before large-scale crop damage occurs. The main advantages are multiple disease categories with high precision, fast results that help immediate decision-making, and reduced expert intervention at the site because of automated classification.

7.Cloud Upload

Once the data from sensors and image processing is ready, the ESP-32 will upload it to the cloud for long-term storage and remote accessibility. This cloud integration enables access to historical trends by farmers and researchers at any time and from any location. It makes sure the data is kept safe, in sync, and accessible. It provides real-time monitoring and enables intelligent automation features such as alerts or recommendations. The cloud acts as a central database for the whole system.

8.Web Dashboard

The Web dashboard is the user interface where all the processed data is shown in an organized format. The devices themselves monitor the soil parameters, the alerts regarding disease, and every system notification. The dashboard will facilitate easier access to data through structured, interactive presentation formats. Farmers can make better decisions with much more ease without needing to be technically savvy. Its remote accessibility feature can allow it to be viewed by the farmer even when he is away from the field.

9.Data Visualization

It is presented by converting raw numeric values into understandable insights with the help of graphs, charts, icons, and alerts that make quick interpretation of trends in soil moisture, plant health, and disease progression easier. Visual representation reduces confusion and represents all complex data in an easy-to-understand form. It enables quicker decision-making by showing obvious patterns and anomalies. Good visualization is now essential in precision farming and monitoring extensive agricultural fields.

10.Management Recommendation

Based on data analysis, the system generates intelligent management recommendations oriented to the needs of the crop. These may include irrigation timing, dosage of fertilizer, treatment procedures in case of disease, or methods of soil correction. By automating such insights, farmers get access to expert-level guidance without requiring deep technical knowledge. Recommendations minimize errors, conserve resources, and enhance productivity. This feature is key to enabling sustainable and data-driven farming.

3.2 Parameters used:

Parameter	Sensor Range (mg/kg)	Justification
Nitrogen (N)	0 – 1000 mg/kg	Covers low to high fertility zones
Phosphorus (P)	0 – 500 mg/kg	Useful for soils with both low and amended P
Potassium (K)	0 – 1000 mg/kg	High K needed for coconut productivity
PH	5.5 to 6.5(red) 6 to 7(black)	acidic more absorption of nutrients leads to toxicity, basic less absorption of nutrients leading to deficiency
Moisture	High Moist Soil 300–600 Moderate Moisture 600–750 Low / Dry Soil 750–1023	Ideal for coconut trees Acceptable Irrigation needed

Fig 3.2 Table of Parameters

The table summarizes the key soil parameters required for healthy coconut cultivation and the corresponding sensor ranges used for monitoring them. Nitrogen (0–1000 mg/kg) and potassium (0–1000 mg/kg) represent the broad fertility levels to ensure that both nutrient-deficient and nutrient-rich soils are adequately measured, while phosphorus (0–500 mg/kg) reflects the typical ranges expected in natural and amended soils. The pH values reflect how soil acidity or alkalinity can affect nutrient absorption—acidic soils lead to toxicity because of excessive nutrient uptake, whereas alkaline soils lead to a restriction in absorption, promoting nutrient deficiencies. Moisture levels are categorized into high, moderate, and low ranges of 300–1023, helping determine if the soil is ideal, acceptable, or needs irrigation. These parameters put together will help farmers make informed decisions to maintain optimum soil health toward coconut productivity.

3.3 Nutrients Comparison:

NUTRIENTS	N	P	K
USES	Nitrogen is necessary for the growth of a plant with good leaf growth and color.	Phosphorous promotes root formation and flower and fruit development	Potassium helps in transfer of water and nutrients in plants and helps photosynthesis.
EXCESS	Over growth with decrease in ability to fruit and flowers, attracts insects and reduces the **strength of stem.**	**Reduces ability** to absorb Zinc and Iron and few **micronutrients** even though they are present in soil.	Leads to **inability to absorb major nutrients** like nitrogen, magnesium and calcium which leads to their deficiency.
DEFICIENCY In Coconut tree	Yellowing of leaves	Nicrosis in younger coconut leaves	Nicrosis and curling

Fig 3.3 Table of Nutrients Comparison

The table highlights the essential roles of nitrogen, phosphorus, and potassium in the growth of coconut plants, together with the effects of their excess and deficiency. Nitrogen supports good development of leaves and their coloration, though its excess causes excessive vegetative growth and attracts pests and weakens the stem. Phosphorus is important for good root development and promotes better flowering and fruiting; its excess interferes with the uptake of micronutrients such as zinc and iron. Potassium helps in the maintenance of water movement, nutrient transport, and photosynthesis; an oversupply decreases the ability of the plant to take up key nutrients such as nitrogen, magnesium, and calcium. Deficiency symptoms in coconut trees involve yellowing of leaves due to lack of nitrogen, necrosis on young leaves due to phosphorus deficiency, and curling or browning of leaf edges due to potassium deficiency. Understanding the behavior of these nutrients helps in maintaining proper soil fertility and healthy growth of coconut.

3.4 Circuit Diagram:

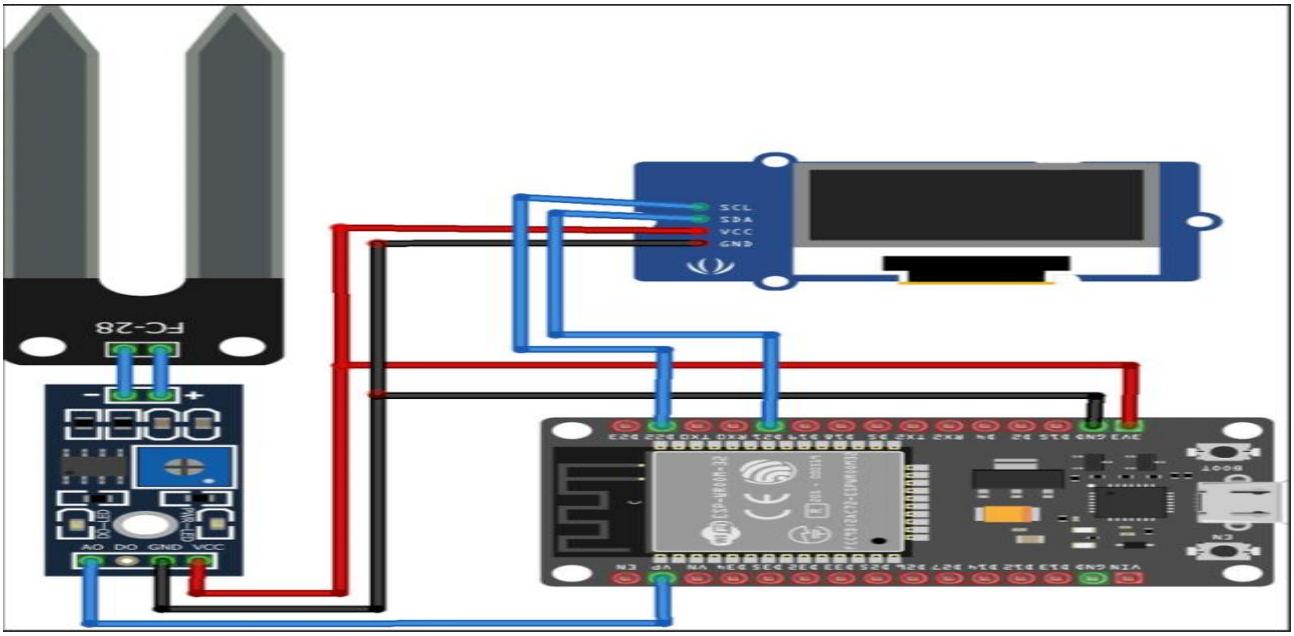

Fig 3.4 Circuit Diagram for Soil Monitoring system

The circuit comprises an ESP32 microcontroller interfacing with a soil moisture sensor, a DHT11 temperature–humidity sensor, and an OLED display. The soil moisture probe is inserted into the soil and connected to its signal conditioning module, which provides an analog output to the ESP32

for measuring the moisture. A DHT11 sensor interfaces via a digital GPIO and provides the data for temperature and humidity. Both sensors are powered from the 3.3V and GND pins of the ESP32. The OLED display is interfaced via an I2C interface using SDA and SCL pins, thus allowing real-time display of sensor readings. All have a common ground to ensure stability in the operation. Further, ESP32 processes the incoming data and shows the values of moisture, temperature, and humidity on the OLED display. Thus, this completes the circuit, which forms a compact smart agriculture module that can monitor the soil condition and environmental parameters simultaneously.

3.5 Hardware Requirements

- ESP32/ESP8266 Microcontroller
- DHT22 Sensor
- Soil Moisture sensor
- Power Supply

3.5.1 ESP32/ESP8266 Microcontroller

Fig 3.5.1 Esp32 Microcontroller

ESP32 and ESP8266 are tiny, smart chips that bring Wi-Fi and Bluetooth (for ESP32) features to everyday electronics, making them perfect for modern projects like smart farming, home automation, and DIY gadgets. Imagine needing an affordable device that connects easily to the internet, reads sensor data, or triggers lights these microcontrollers make it simple without big, expensive equipment. ESP8266 is known for being easy to use and cost-effective, suited to simple Wi-Fi projects or basic monitoring.

ESP32 goes much further: it's fast, powerful, and handles more tasks at once (dual-core processor), with Bluetooth support and extra pins, so you can connect more sensors and do more sophisticated activities at the same time. Both chips help users automate and control their environment remotely, with the ESP32 being especially helpful for complex or demanding tasks where higher speed or more connections are needed. This flexibility means anyone from hobbyists to professional engineers can build smart solutions quickly and affordably.

3.5.2 DHT22 Sensor

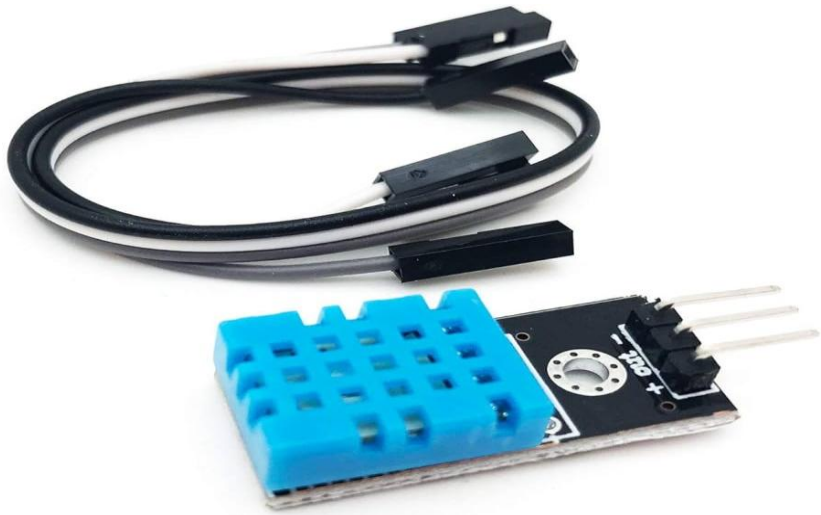

Fig 3.5.2 DHT22 sensor

The DHT22 sensor is a simple, affordable device that measures both temperature and humidity in the air. Think of it as a tiny weather station that tells you how warm or cool it is and how much moisture is in the air around you. It uses a special part that changes its electrical resistance depending on humidity like how your clothes feel stickier on humid days and a tiny thermometer inside to check the temperature. The sensor then sends this information in a digital signal to a microcontroller like an ESP32 or Arduino, so the data can be read, displayed, or used to control devices like fans or irrigation systems. Because it's easy to use and accurate, the DHT22 is popular in projects like home automation, weather stations, and smart agriculture, helping people keep track of environmental conditions in a practical way.

3.5.3 Soil Moisture sensor

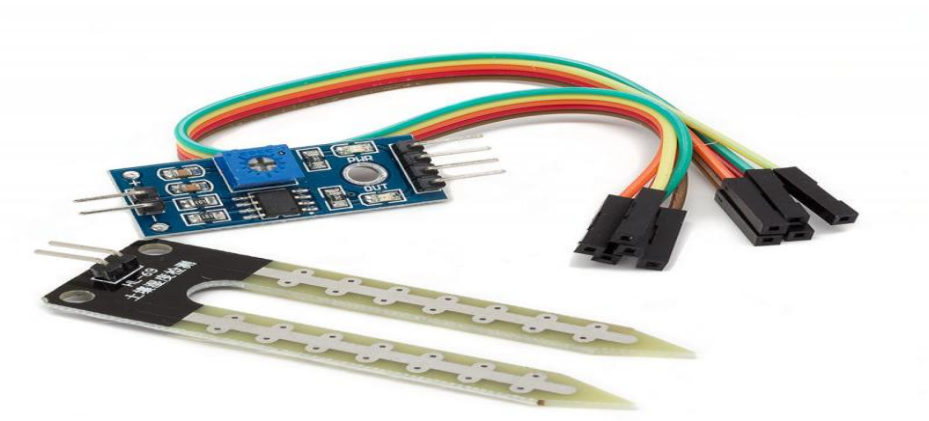

Fig 3.5.3 Soil moisture sensor

Soil moisture sensors are essential tools primarily used in agriculture to measure the water content in soil, providing real-time data that helps farmers optimize irrigation. By knowing exactly how much moisture the soil holds, farmers can prevent overwatering or underwatering, saving water and improving crop health. These sensors enable precise irrigation scheduling, leading to better resource efficiency, higher crop yields, and reduced water wastage. Beyond irrigation, soil moisture sensors are also used in precision agriculture to manage fertilizer application and monitor soil health. They play a significant role in drought monitoring and environmental protection by helping detect soil moisture changes that could signal risks such as soil erosion or landslides.

Additionally, these sensors support scientific research in plant growth and climate studies and are used in landscaping and turf management to maintain optimal soil moisture levels. Modern technology, including satellite-based sensors, further enhances the capability to monitor soil moisture remotely across large or difficult-to-access areas. Overall, soil moisture sensors provide accurate, timely soil condition information that supports sustainable farming practices, conserves water resources, and improves agricultural productivity while reducing labor and costs.

3.5.4 Power Supply

Fig 3.5.4 Power Supply

The ESP32 needs a steady 3.3V power supply to work properly, and it usually requires at least 500mA of current to handle its operations smoothly. You can power it through a USB 5V input, as the board has an onboard voltage regulator that steps it down to 3.3V safely. Alternatively, it can be powered directly with a regulated 3.3V supply, but make sure the voltage stays within the safe range to avoid damaging the module.

3.6 Software Requirements

- Development Tools
- Frontend Stack service
- Backend Services

3.6.1 Development Tools

3.6.1.1 Node.js v16.0 or higher :

Fig 3.6.1.1 Node.js software

Node.js v16 and higher versions have transformed the way developers build applications by making the process faster, smoother, and much more efficient. Imagine a tool that allows programmers to write JavaScript not just for websites, but also to run servers and handle multiple tasks at the same time without slowing down that's Node.js. This version brought new upgrades like faster performance through an improved JavaScript engine, better security features, and streamlined ways to handle asynchronous tasks, which means it can manage many users' requests without making anyone wait. It's like having a super-efficient manager who keeps everything running seamlessly in the background. Because it uses the same language on the front end and back end, developers find it easier to build entire web applications quickly and cohesively. Plus, it supports real-time applications such as chat services and live updates, which are crucial in today's connected world. All these improvements make Node.js v16 not just a tool but a vital part of creating modern, scalable, and responsive applications that users enjoy interacting with every day. In short, it empowers developers to deliver powerful, reliable, and fast experiences that feel natural and engaging to people using the web and apps around them.

3.6.1.2 npm v8.0 or higher :

npm version 8 and higher have become an essential companion for developers working with JavaScript and Node.js. Imagine it as a smart assistant that helps you manage all the code packages and libraries your project needs, making your work easier and more organized. This version introduced handy features like improved handling of workspaces, which means you can manage multiple related packages in one

project more smoothly. It also brought in better tools for debugging, making it simpler for developers to find and fix issues quickly. The ability to override package versions helps prevent conflicts, so your project stays stable and reliable. Because npm manages dependencies behind the scenes, developers can focus more on writing great code instead of worrying about compatibility or installation hassles. Overall, npm 8 is like a trusted helper that keeps your development process efficient, secure, and hassle-free, playing a vital role in bringing complex applications to life seamlessly.

3.6.1.3 Code editor (VS Code recommended):

Visual Studio Code, or VS Code, is much more than just a tool to write code-it's like having a helpful and intuitive workspace that makes coding feel natural and even enjoyable. It’s designed to be fast and lightweight so that you can dive straight into your projects without any overhead slowing you down. What really makes VS Code stand out is how smart it is; it feels like the editor understands your code, offering helpful suggestions, pointing out mistakes as you write, and even helping you fix them quickly. Plus, it supports almost every programming language and can be tailored to fit your specific needs, thanks to a vast library of extensions. It’s also great for teamwork, as it integrates tools for version control and collaboration seamlessly, making it easier to share and improve code with others regardless of where they are. In essence, VS Code turns the sometimes-daunting task of coding into a smoother, more productive, and creative experience, helping developers focus on what truly matters-bringing ideas to life with less friction and more confidence.

3.6.2 Frontend Stack service

3.6.2.1 React 19.2.0:

React 19.2 brings a fresh way to make your web apps feel super smooth and effortless. One of its coolest features is the new <Activity> component, which acts like a smart organizer for your app’s parts. Instead of constantly loading and unloading pieces of your interface-which can slow things down, react 19.2 lets you keep parts running quietly in the background, preserving their state without draining resources. Imagine you’re switching between tabs or pages, and everything feels instant because the app already remembers where you left off, without reloading everything from scratch. This makes apps not only faster but also more enjoyable to use, especially when internet connections are shaky. Plus, it helps developers keep their code neat and less buggy with smarter handling of background tasks. It’s

like having an expert assistant that knows exactly what to prioritize, making sure your app always performs at its best, effortlessly and gracefully, so users can just focus on what matters most.

3.6.2.2 React Router DOM 7.9.3

React Router DOM 7.9.3 is like a helpful guide that makes moving around your web app easy and smooth. It connects different parts of your app to specific URLs, so users can click links and see new pages instantly without waiting for the whole site to reload. This version improves how developers organize and protect pages, keeping sensitive info safe and making sure the app feels seamless. It also helps handle data loading in the background, so everything feels quick and natural. With React Router DOM 7.9.3, navigating your app becomes effortless and enjoyable, both for users and developers.

3.6.2.3 Chart.js 4.5.0:

Chart.js 4.5.0 is a powerful and easy-to-use tool that helps you create beautiful, animated charts right on your website. It supports many different chart types like line, bar, pie, and bubble, giving you plenty of ways to show your data clearly. What makes it special is how customizable it is, so you can tweak colors, shapes, and animations to fit your style perfectly. It’s designed to work smoothly across all modern browsers, ensuring your charts look great everywhere. Plus, with its simple setup, even beginners can start making interactive charts quickly. In short, Chart.js 4.5.0 turns boring numbers into engaging visual stories that anyone can understand at a glance.

3.6.3 Backend Services

3.6.3.1 Firebase Authentication:

Firebase Authentication is a simple and secure way to manage user sign-ins for your app. It supports many sign-in options, like email, phone numbers, and popular social accounts such as Google and Facebook, so users can pick whichever they feel comfortable with. This service handles all the tricky details of verifying users, keeping their data safe, and managing passwords, so you don’t have to build this complicated system from scratch. It’s especially important because it helps protect user accounts and makes signing in smooth and easy, improving the overall user experience. Plus, it integrates well with other Firebase tools, helping developers build apps faster and with fewer worries about security. In

short, Firebase Authentication is like a trusted gatekeeper that keeps your app secure while making it simple for users to access what they need.

3.6.3.2 Firebase Realtime Database:

Firebase Realtime Database is like a magical notebook that instantly syncs your app's data across all users' devices in real time. Whenever one person updates something, everyone else sees the change without refreshing or waiting, making it perfect for chat apps, live collaboration, or games. It even works smoothly offline, saving changes locally and syncing them once the connection is back. This makes your app feel fast, reliable, and always up-to-date no matter where users are. It also keeps data safe and secure with strong rules and integrates easily with Firebase Authentication to control who can see or change what. In short, Firebase Realtime Database helps developers build engaging, real-time experiences that users love, with less hassle behind the scenes.

3.6.3.3 Firebase Storage:

Firebase Storage is a simple and powerful tool that lets you store and serve user-generated files like photos and videos directly from your app. It uses Google's secure cloud infrastructure, so your files are safe and accessible anytime, anywhere. What makes it special is how easily it integrates with Firebase Authentication, so only the right users can upload or access their files. It's built to handle all the tricky parts like handling slow or interrupted networks by resuming uploads automatically. Firebase Storage scales with your app, so whether you have a few users or millions, you won't have to worry about running out of space or slowing down. In short, it helps developers effortlessly manage and protect files while providing a smooth experience for users.

3.7 Implementation:

This figure shows an intelligent system designed to monitor and manage crop health and soil conditions, making life easier for farmers and growers. The heart of this setup is the ESP-32 Dev kit, which receives power from a reliable supply and connects to a soil moisture sensor to keep tabs on water levels in real time. The results are displayed instantly on a handy OLED screen for quick checks. Data from the ESP-32 is uploaded to the cloud, so farmers can see it on a web dashboard wherever they are. This dashboard lets users visualize important information including soil moisture and disease status with charts and clear graphics, helping them make informed decisions.

Meanwhile, images such as those of crops go through preprocessing and disease detection algorithms that analyze NPK (nutrient) and pH values. Results are classified for diseases before being sent to the cloud, allowing instant updates and trends on the dashboard. Finally, the system provides personalized management recommendations based on the data, guiding farmers to take the right actions without guesswork. Overall, this combination of smart sensors, cloud connectivity, and intelligent recommendation helps save time, reduce risk, and enables anyone to care for their crops more confidently and efficiently.

3.7.1 Firestore Collections (user collection):

```
users/{userId}
{
firstName: string,
lastName: string,
email: string,
phoneNumber: string,
location: string,
farmSize: number,
createdAt: timestamp,
lastLogin: timestamp
}
```

Purpose: To store and manage user information such as name, email, phone number, and farm details.

Steps:

- When a new user registers, generate a unique userId.
- Collect the following user details: firstName, lastName, email, phoneNumber, location, and farmSize.
- Store these details at users/{userId} in Firestore.
- Add the createdAt timestamp to record the time when the user joined.
- Update the lastLogin timestamp whenever the user logs in.
- If the user changes profile details, update that particular field.
- Read this collection whenever user information is needed for the app.
- Display user information on the dashboard or profile screen.

3.7.2 Analyze History Collection:

Json

analysisHistory/{docId}

```
{
userId: string,
analysisType: string, // 'soil', 'plant', 'combined', 'comprehensive'
timestamp: timestamp,
// Optimized Analysis Data
analysisData: {
summary: string,
confidence: number,
keyFindings: array,
recommendations: array,
riskLevel: string
},
// Image References
images: {
soilImageUrl: string,
plantImageUrl: string,
diseaseImageUrl: string
},
// Sensor Data (for comprehensive analysis)
sensorData: {
temperature: number,
humidity: number,
soilMoisture: number,
timestamp: timestamp
}
}
```

Purpose: To save soil, plant, or combined analysis reports for future reference.

Steps:

- Assign a new docId for a user uploading images or requesting analysis.
- Store the userId to connect the analysis to the right account.
- Store the type of analysis - soil / plant / combined.
- Add current timestamp of when analysis is done.
- Run the analysis model AI/ML and create:
  - summary
  - confidence score
  - key findings

- recommendations
- riskLevel

- Save these under analysisData within the document.
- Upload images of soil, plants and disease. Store their URLs under images.
- If the analysis is comprehensive, collect also sensor data (temperature, humidity, soil moisture), storing it under sensorData.
- Pull these records whenever the user views past analysis reports.

3.7.3 Realtime Database Structure:

```
json
{
"LatestData": {
"temperature": 27.5,
"humidity": 65,
"soilMoisture": 45,
"lastUpdated": "2025-10-10T10:30:00Z"
},
"SensorData": {
"timestamp_1": {
"temperature": 27.5,
"humidity": 65,
"soilMoisture": 45,
"timestamp": "2025-10-10T10:30:00Z"
},

"timestamp_2": {
"temperature": 28.0,
"humidity": 63,
"soilMoisture": 42,
"timestamp": "2025-10-10T11:30:00Z"
}
},
"Thresholds": {
"temperature": {
"min": 20,
"max": 35,
"optimal": 27
},
"humidity": {
"min": 60,
"max": 80,
"optimal": 70
},
```

```
"soilMoisture": {
"min": 40,
"max": 70,
"optimal": 55
}
}
}
```


Purpose: To monitor real-time sensor readings and store live agriculture data.

A. For “LatestData” Node

- Read live sensor values from ESP32 or IoT module.
- Update the “LatestData” section with :
  - Temperature
  - humidity
  - soilMoisture
  - lastUpdated timestamp
- Replace old values every time new data arrives.

B. For "SensorData" Node (Historical Storage)

- Create a unique timestamp key when new values of sensors arrive.
- Store the sensor values at "SensorData/timestamp_x" including its own timestamp.
- This builds a history of sensor readings for graphs or trend analysis.

C. For “Tihresholds” Node

- Predefine min, max and optimal values for:
  - Temperature
  - humidity
  - soilMoisture
- Save these values under the “Thresholds” node.
- Compare the thresholds against the sensor data whenever new data arrives.
- If data goes below/above limits, trigger warnings or notifications.

3.8 Model Development

3.8.1 Registration Process

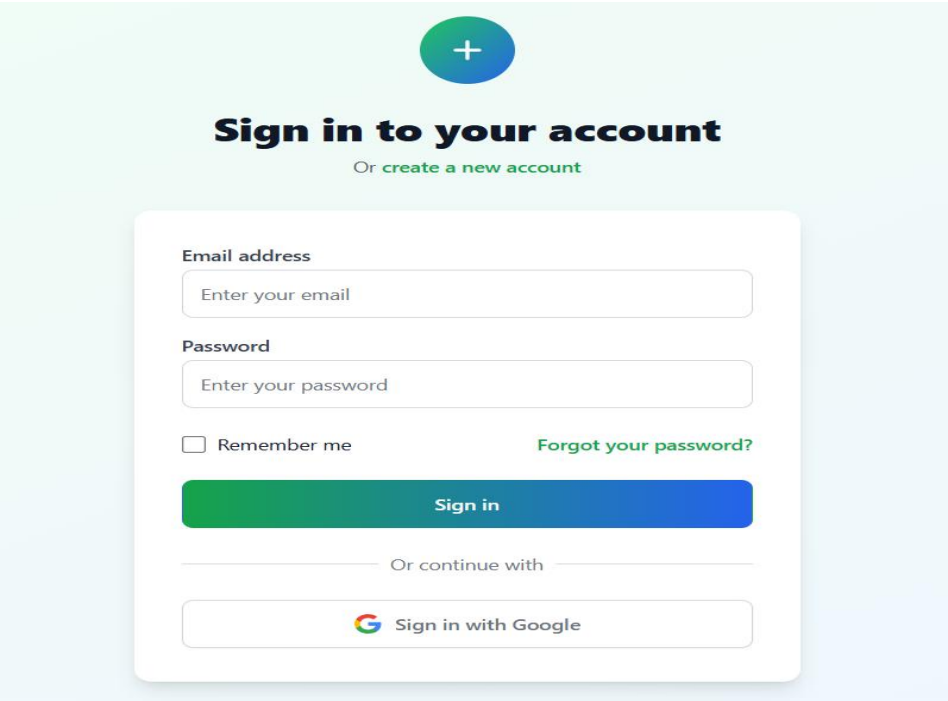

Fig 3.8.1 Registration Process

- Email validation.
- Password strength checking.
- User profile creation in Firestore.
- Welcome email (optional).
- Automatic login after registration.

3.8.2 Dashboard Monitoring

This dashboard is like having a friendly assistant by your side, helping you keep an eye on your coconut plants with just a quick glance. It greets you warmly and instantly shows how many plants are being looked after, how many health checks have been run that day, and whether there's anything urgent that needs your attention. The live sensor data feels almost like a conversation with your garden, telling you right away if things are a bit too hot, a little too dry, or if your plants are thirsty and need water. With every update, it feels reassuring to know exactly how your plants are doing, making it simple to respond to their needs and keep them thriving.

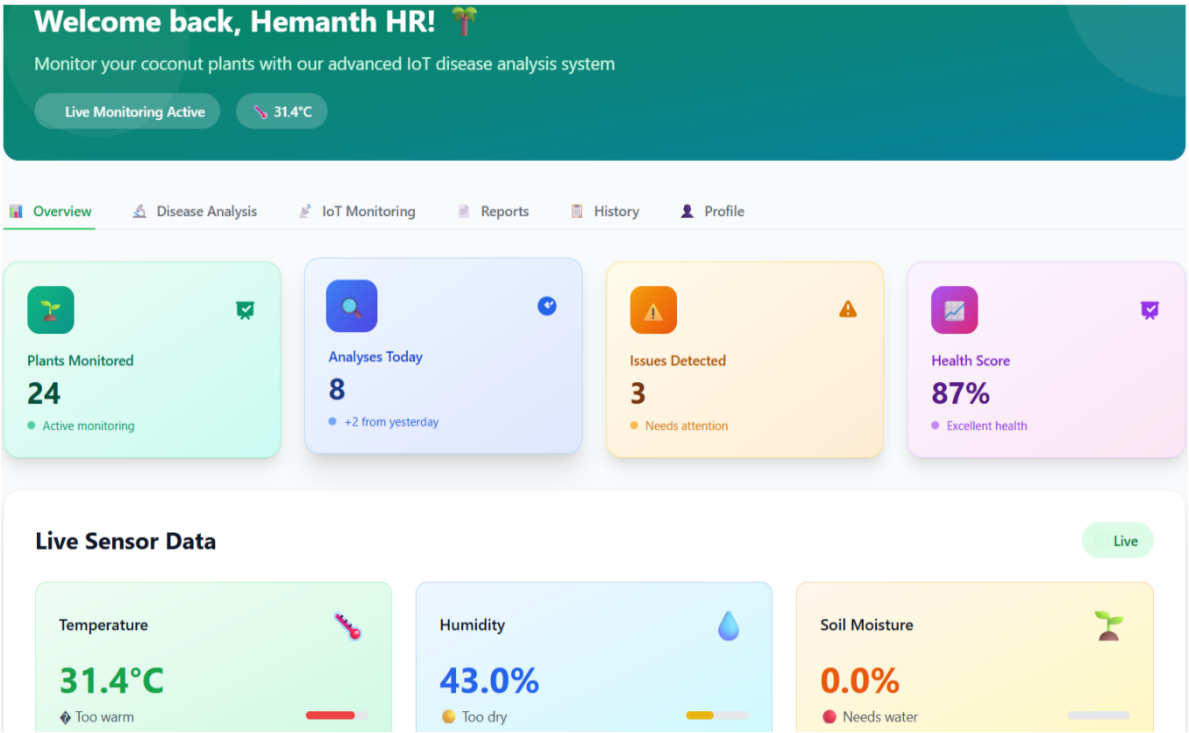

Fig 3.8.2 Dashboard Management

If any problems pop up, you see clear alerts, so nothing slips through the cracks even minor issues get noticed and dealt with before they grow. The health score, shining at 87%, brings reassurance, letting you know that your plants are generally in excellent shape and well cared for. With every message and metric, the dashboard makes plant care simple, supportive, and much more personal.

3.8.3 Disease Analysis Monitoring

The AI-Powered Analysis Center is all about making farming smarter, easier, and more reliable, especially for people who work with crops and soil every day. This friendly digital hub gives farmers and growers tools to check their soil's nutrients, spot plant diseases early, and even blends powerful AI insights with real-time IoT sensor data for a truly modern touch. Instead of guessing, users can learn exactly what their soil or plants need, get advice tailored for their farm, and make confident decisions that save time and resources. The whole platform is designed to feel helpful, approachable, and supportive like having a team of expert advisors right at your fingertips, ready to guide you toward healthier crops and better harvests, every single season.

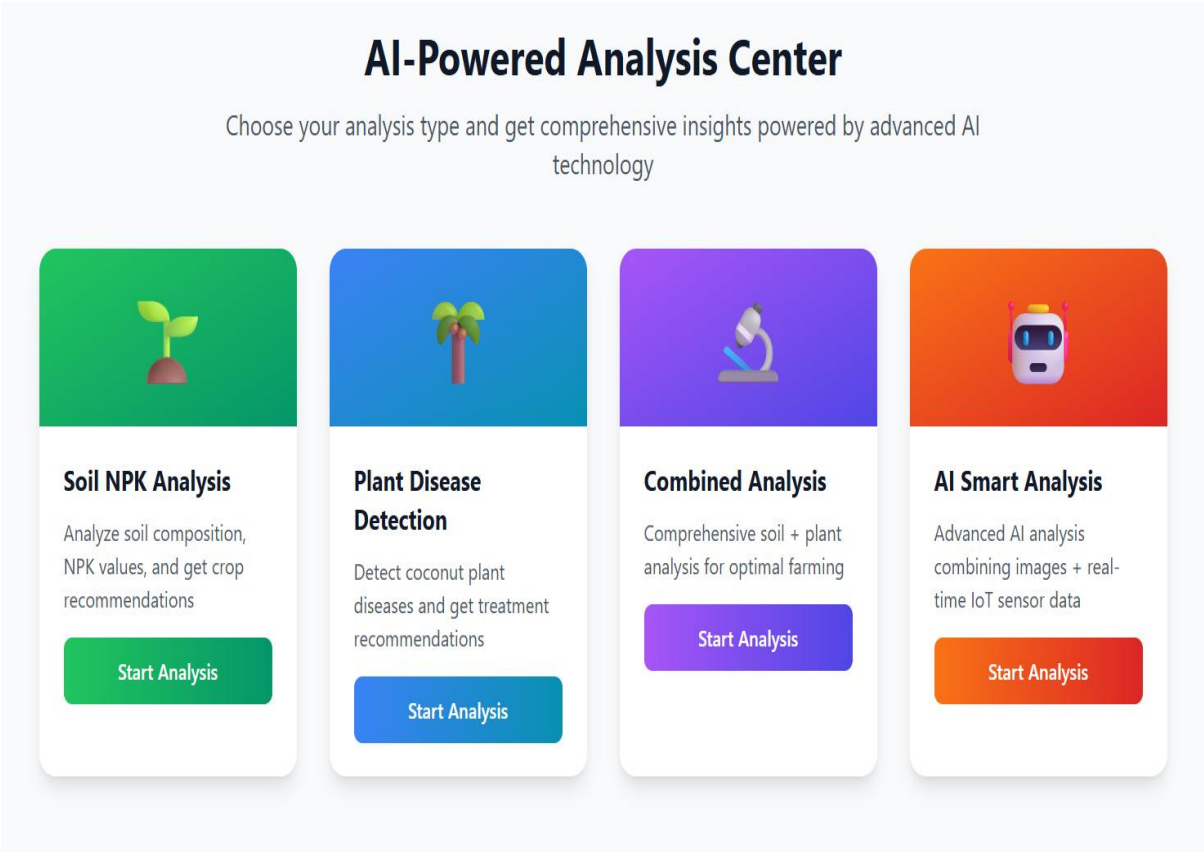

Fig 3.8.3 Disease Analysis Management

3.9 Complete list of common coconut diseases, grouped into understandable categories:

1. Leaf Spot Disease

Caused by fungi such as Helminthosporium and Pestalotiopsis that produce brown to black spots on leaves.

2. Grey Leaf Spot

Shows greyish oval patches that weaken the leaves and reduce photosynthesis.

3. Leaf Blight

Leaves turn brown or black from the tip down, usually as a result of fungal infection.

4. Yellow Leaf Disease

Leaves gradually turn yellow and droop; often related to nutrient problems or infection.

5. Lethal Yellowing Disease (Leaf Phase)

Yellowing of older leaves is an early symptom, eventually the whole palm declines.

6. Anthracnose of Coconut Leaves

Caused by Colletotrichum fungus; produces irregular, dark brown patches.

3.10 The Optimum Environmental Ranges For Coconut Cultivation (Temperature, Humidity, And Soil Parameters).

1. Temperature Range

- Optimum: 27°C – 32°C
- Minimum required: 20°C
- Maximum tolerance: 38°C
- The coconut palm thrives well in warm, humid tropical conditions.

2. Range of Humidity

- Optimum Relative Humidity: 70% – 80%
- High humidity favors good flowering, nut formation, and healthy foliage.

3. Soil Moisture

- Optimum soil moisture: 55% – 70%
- Soil should be moist but well-drained, without waterlogging.

4. Soil pH

- Optimum pH: 5.5 – 7.0
- Slightly acidic to neutral soil is optimum.

5. Type of soil

Best soils:

- Loamy soil

- Red sandy loam
- Alluvial soil
- Coastal sandy soil, with proper irrigation

The soil needs to have:

- Good drainage
- High water-holding capacity
- No hardpan layers

6. Rainfall Requirement

- Optimum annual rainfall: 1500 – 2500 mm
- Evenly distributed throughout the year.

CHAPTER 4

RESULT

4.1 Hardware Component Model

This image shows a simple electronics project setup, likely for a DIY or educational application involving sensors and microcontrollers. On the white paper and breadboard, different components are connected using jumper wires. You can see a microcontroller module (possibly an ESP32 or ESP8266), a blue sensor (which looks like a DHT11 or DHT22 for temperature and humidity), and another sensor unit nearby. This setup brings together the building blocks of modern electronics: sensing, processing, and connecting data, all in a hands-on way that feels approachable and fun, even for beginners. Creating circuits like this is a great way to learn how smart devices work and how they interact with the physical world, making technology feel a little more magical and a lot more understandable.

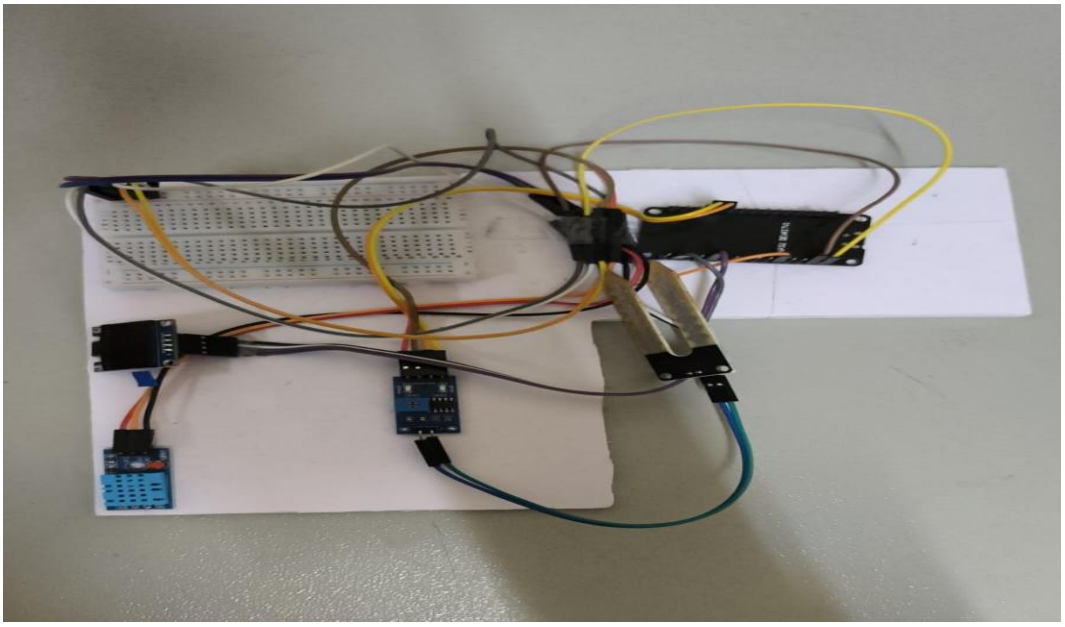

Fig 4.1 Hardware model

4.2 Analyse soil composition with NPK values

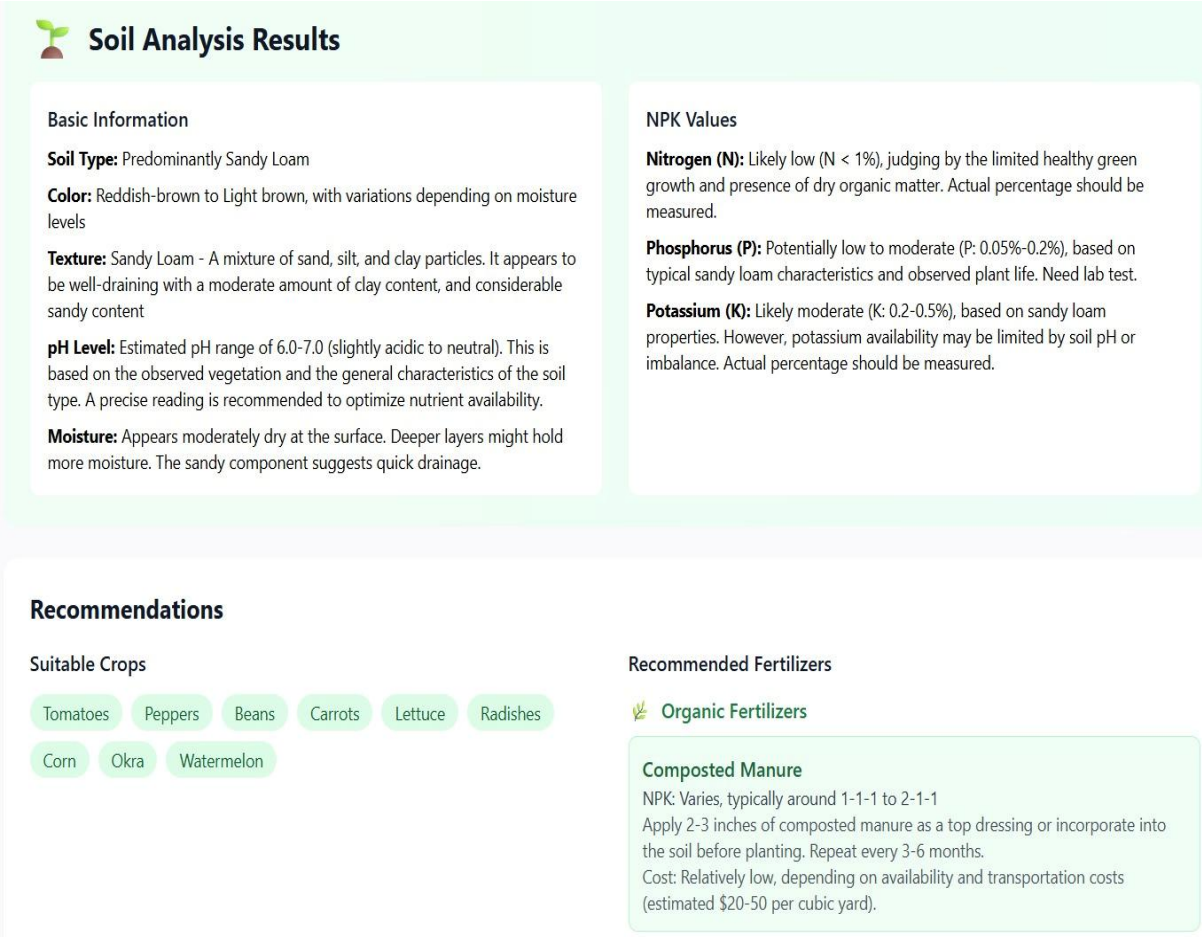

Fig 4.2 Soil Analysis with NPK values

The report above gives a clear and friendly picture of your soil's health, using language that's easy to understand. Your soil is described as red lateritic, with a loamy texture that's slightly acidic and moderately moist although the surface is starting to dry out just a bit. The nutrient levels, especially nitrogen and phosphorus, are quite low, while potassium is in the medium range. Based on this, the system recommends a variety of crops like cashew, mango, jackfruit, pineapple, tapioca, and several others that will thrive in these conditions.

To help your crops do their best, the report also suggests organic fertilizers such as composted cow manure and vermicompost. It gives you practical details about how much to use, how often, and even includes the approximate cost per kilogram. This approach not only takes the guesswork out of farming but also encourages eco-friendly practices. With this kind of insight, you can feel more confident about what to grow and how to care for your land, all while making informed choices that are good for both your crops and the environment.

4.3 Disease Detection and recommendations(leaf)

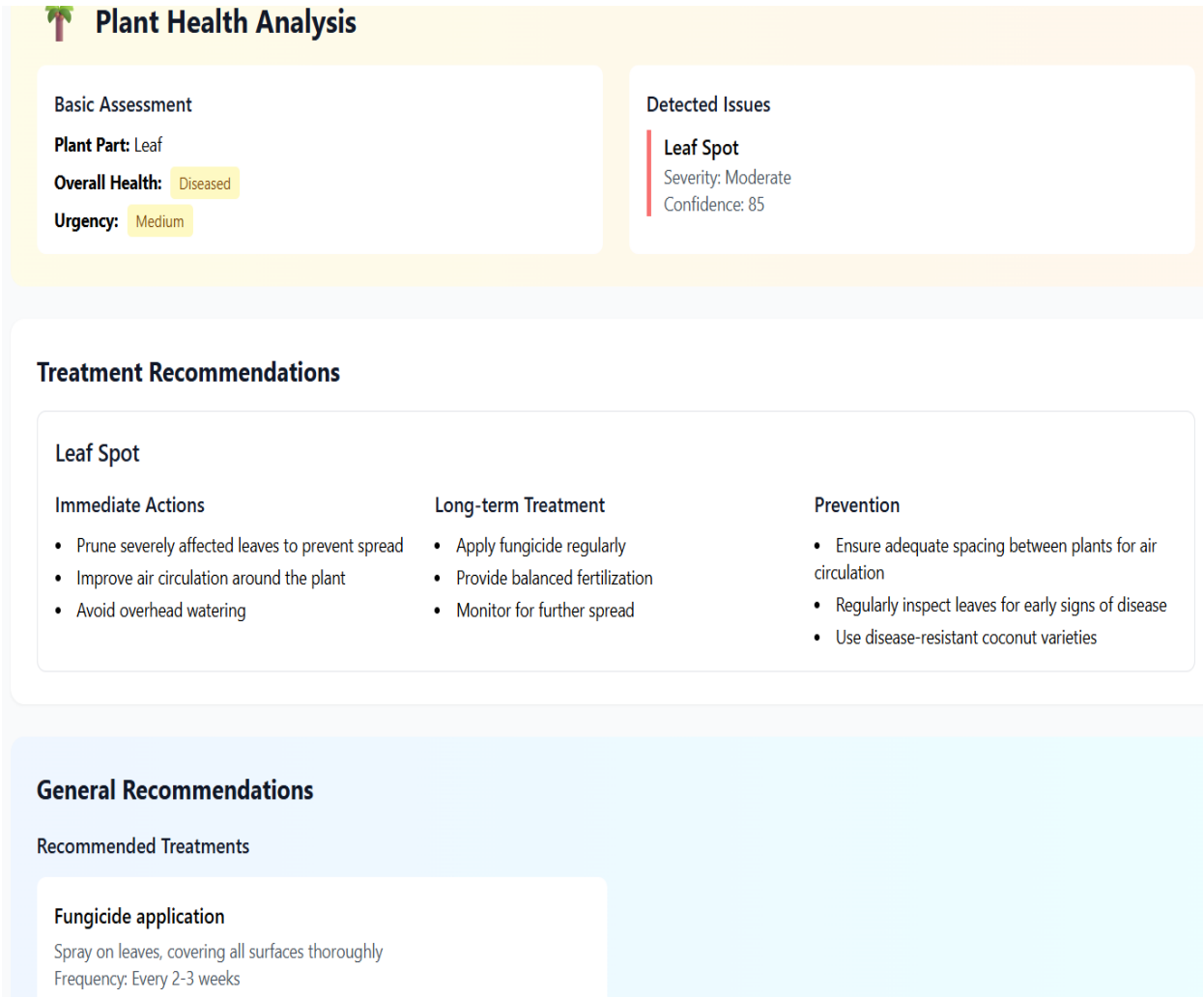

Fig 4.3 Disease detection (leaf)

This plant health report gives you a caring, straightforward look at what's happening with your crop's leaves. It points out that the leaves are showing signs of disease specifically, a moderate leaf spot problem. The urgency is marked as medium, which means it's important but not an emergency. To help you take action, the report suggests pruning away badly affected leaves, improving airflow, and skipping overhead watering right away.

For longer-term care, it recommends using fungicides regularly, maintaining balanced fertilization, and keeping a close watch for more signs of spread. Preventing future problems is also part of the advice: make sure your plants have space to breathe, check leaves often for early signs of disease, and consider planting disease-resistant coconut varieties. With easy-to-follow guidance like spraying fungicide every 2-3 weeks, this analysis helps you feel empowered to address issues quickly, protect healthy plants, and stay one step ahead of disease so your crops stay vibrant and strong.

The nutrient levels, especially nitrogen and phosphorus, are quite low, while potassium is in the medium range. Based on this, the system recommends a variety of crops like cashew, mango, jackfruit, pineapple, tapioca, and several others that will thrive in these conditions. To help your crops do their best, the report also suggests organic fertilizers such as composted cow manure and vermicompost. It gives you practical details about how much to use, how often, and even includes the approximate cost per kilogram. This approach not only takes the guesswork out of farming but also encourages eco-friendly practices. With this kind of insight, you can feel more confident about what to grow and how to care for your land, all while making informed choices that are good for both your crops and the environment.

4.4 Disease Detection and recommendations(caterpillar)

The plant health analysis above focuses on a corn stalk suffering from stem borer infestation, which is a common and damaging pest for maize. Stem borers, when they attack, often leave clear signs such as dead heart (where the central shoot dries up), small bore holes on stems, tunnels inside the stalk, stunted growth, and wilting. As the larvae burrow into the stem and feed on its tissues, these symptoms become more severe sometimes leading to extensive tunneling, significant wilting, and weakened stalks that may even fall over. Early-stage damage includes the presence of "shot holes" on leaves, and as the pest population grows, adult moths and pupae can appear within the plant, indicating a well-established infestation that needs attention soon.

To support farmers in managing this problem, the analysis offers practical steps divided into immediate actions, long-term solutions, and preventive strategies. Immediate recommendations include removing the visible caterpillars, cutting back extensively damaged portions, and isolating infested plants to

limit spread. For long-term control, using recommended insecticides, promoting natural predators like wasps, practicing crop rotation, and maintaining good crop sanitation by clearing plant debris are targeted strategies. Encouraging beneficial insects, monitoring for signs of infestation, and using resistant varieties if available, help prevent future outbreaks, making the guidance both effective and sustainable for keeping corn crops healthy.

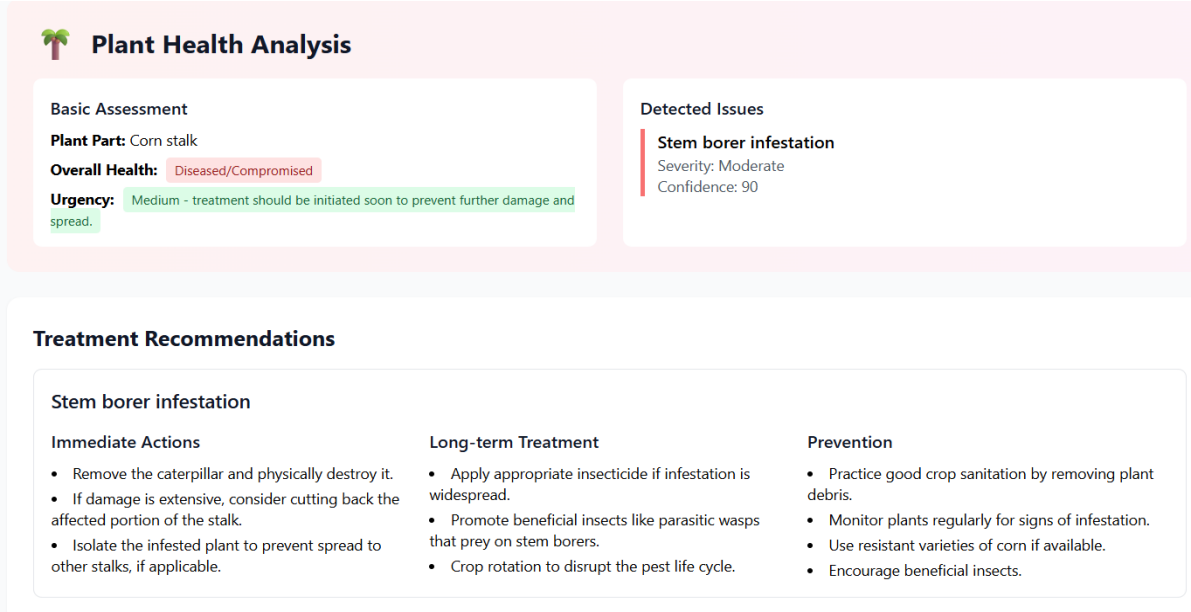

Fig 4.4 Disease detection (caterpillar)

4.5 Combined soil analysis and Disease Detection

The assessment shown in the figures feels reassuring for coconut growers. It highlights that the soil is likely sandy loam with a good dose of organic content, giving coconut trees the drainage they need while helping roots thrive. Although nutrients like nitrogen and phosphorus seem present, the recommendation is to double-check those critical levels especially potassium because coconut trees depend on them for long-term health and fruiting.

Looking at the coconut plant itself, things appear positive. The green, flowering fronds suggest a healthy and productive tree at this stage. Yet, the advice is practical keep monitoring for any hidden signs of disease and regularly test the soil to ensure every nutrient the plant needs is in balance. With this gentle guidance, even less-experienced farmers can confidently care for their trees, knowing they're supported by thoughtful, farmer-friendly science.

Soil Analysis

Soil Type: Based on the image, the soil appears to be primarily sandy loam or clay loam with a reddish hue. This suggests the presence of iron oxides. The texture appears somewhat compact and possibly lacking in organic matter.
Coconut Suitability: The soil's current condition is likely suboptimal for coconut farming. Sandy loam is generally good for drainage, which is important, but the apparent lack of organic matter and potentially low NPK values need addressing to ensure proper nutrient availability for coconut trees. Amendment with organic matter and proper fertilization are crucial.

NPK Values:

- Nitrogen: Likely low, as evidenced by the lack of lush, green vegetation in the surrounding area.
- Phosphorus: Potentially moderate, depending on the soil's parent material. Further testing is needed for accurate assessment.
- Potassium: Potentially low to moderate. The presence of some green grass suggests some potassium, but coconut trees are heavy feeders, so this could be a limiting factor.

Plant Analysis

Health Status: The coconut frond shows signs of chlorosis (yellowing), especially between the veins. This indicates potential nutrient deficiencies.

Detected Issues:

Likely no specific disease visible in the image, but nutritional deficiencies can weaken the plant and make it more susceptible to disease.

Soil-Plant Correlation

Relationship: The yellowing leaves indicate a potential nutrient deficiency stemming from the soil. The apparent low NPK and organic matter levels in the soil are likely contributing to the plant's poor health. Inadequate nutrient supply from the soil directly impacts leaf color and overall growth.

Nutrient Matching: The soil likely does not adequately supply the necessary nutrients for optimal coconut growth, particularly potassium, nitrogen, magnesium and potentially iron. The plant's symptoms directly correlate with these potential deficiencies.

Fig 4.5 Both soil analysis Disease detection

Integrated Recommendations:

Nutrient deficiencies (K, Mg, N, Fe, Mn)

Soil: Conduct a soil test to confirm nutrient levels. Amend soil with organic matter (compost, manure) to improve soil structure and nutrient retention. Apply appropriate NPK fertilizer and micronutrient supplements (especially magnesium and iron) based on soil test results and coconut tree needs.
Plant: Foliar application of micronutrients (magnesium sulfate, iron chelate) can provide a quick boost. Apply fertilizer around the drip line of the tree. Consider deep root fertilization for faster nutrient uptake.
Timeline: Significant improvement in 3-6 months with consistent treatment.

Poor soil structure and drainage

Soil: Incorporate organic matter to improve soil structure and drainage. Ensure proper drainage to prevent waterlogging, which can exacerbate nutrient deficiencies.
Plant: Monitor the health of roots and remove any dead or decaying material. Apply a fungicide, as needed.
Timeline: Improved soil structure within 6 months; better root health immediately.

Integrated Management Plan

Immediate Actions

- Conduct soil testing to determine precise nutrient deficiencies.
- Apply a foliar spray of micronutrients (Mg, Fe, Mn) to provide a quick boost.
- Begin incorporating organic matter around the base of the coconut tree.

Monthly Plan

- Monitor plant health for changes in leaf color and growth rate.
- Continue adding organic matter to the soil.
- Ensure adequate watering, especially during dry periods.

Seasonal Activities

- During the rainy season, apply a balanced NPK fertilizer after confirming soil test.
- Prune any dead or diseased fronds to improve air circulation and prevent disease spread.
- Consider cover crops to improve soil health and nutrient cycling.

Fertilization Schedule

Month	Fertilizer	Amount	Method
Next Month	Magnesium Sulfate (Epsom salts)	As directed on product label per tree size, usually a few pounds	Apply around drip line of tree or as a foliar spray, diluted in water

Fig 4.6 Integrated Recommendations for Soil analysis and Disease detection.

4.6 Applications, Advantages and Disadvantages

4.6.1 Applications

- Real-time soil monitoring using IoT sensors for efficient farm management.
- Coconut disease detection through image processing and deep learning techniques.
- User-friendly mobile interface for farmers to access soil and crop health data.
- Automated suggestions for disease management and treatment.

4.6.2 Advantages

- Real-time monitoring helps optimize soil conditions, improving crop yield.
- Early detection of coconut diseases reduces crop damage and loss.
- An easy, user-friendly interface it available to farmers who have little technical expertise.
- The system can suggest suitable management techniques, reducing dependency on agricultural experts.

4.6.3 Disadvantages

- Sensors may require periodic calibration and maintenance for accurate readings.
- Initial setup cost and complexity may discourage adoption among small-scale farmers.
- The accuracy of disease detection depends on the quality of captured images and training data. Continuous power and internet connectivity are required for seamless operation and data updates.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

Smart Coco is transforming coconut farming by using real-time monitoring and smart technology to make farming more efficient and sustainable. This system combines sensors, artificial intelligence, and mobile apps to keep a close watch on everything from soil moisture to coconut growth stages. Farmers get timely insights and precise recommendations, helping them save water, improve yield, and reduce disease risks. This technology not only boosts productivity but also makes farming smarter, easier, and more eco-friendly.

In short, Smart Coco empowers coconut farmers with the kind of data and tools that make a real difference in their daily work. By replacing guesswork with accurate, real-time information, it helps farmers make better decisions watering just the right amount, spotting problems early, and harvesting at the perfect time. This approach promises a brighter future for coconut farming by supporting healthier trees, reducing wastage, and increasing profits in harmony with nature.

5.2 Future scope

This project develops an integrated smart agriculture system for coconut farming that combines real-time soil monitoring with image processing-based disease detection to address key challenges in soil health management and early disease identification. The soil monitoring module uses sensors interfaced with a microcontroller to continuously measure moisture, pH, temperature, and nutrient content, providing farmers with real-time data through an easy-to-use interface for informed irrigation and fertilization decisions. The disease detection component employs image processing techniques on coconut tree photographs to analyze leaf patterns and identify infection signs, enabling early disease recognition and prompt farmer intervention to minimize crop losses. Together, these technologies create a comprehensive tool that empowers farmers to manage their fields more effectively and sustainably while improving plant health and yield. Future enhancements include advanced AI algorithms for more accurate disease diagnosis, expansion to monitor other crops, integration with automated irrigation systems, and development of cloud-based analytics with mobile apps for global farmer accessibility.

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